Geometric Spanning Trees that Minimizes the Wiener Index

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Wiener Index in Graphs

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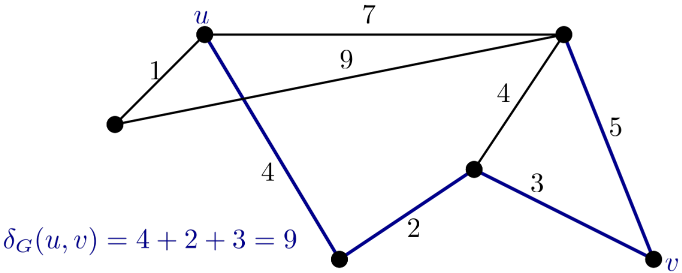

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- The MRCST problem is NP-Complete. There exists a PTAS for MRCST.

Problem Statement: Geometric Spanning Trees

Reference: WADS 2023 [Abu-Affash et al., 2023]

1. Input: A set P of n points in the plane.
2. Goal: Construct a **spanning tree** on P that minimizes the **Wiener index**.
3. The weight function is defined as the Euclidean distance between points.

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- The hamiltonian path of P that minimizes the Wiener index is not necessarily planar.
- Computing such a hamiltonian path is NP-Hard.
- There exists a $(1 + \varepsilon)-$Approximation algorithm for finding a Hamiltonian path that minimizes the Wiener index.*

* This relies on distortion embeddings and is overall complicated to implement.

Divide and Conquer Algorithm for Approximating Hamiltonian Paths

Algorithm 1 DivideAndConquerWiener(P)

procedure DivideAndConquerWiener(P, $depth = 0$, $maxDepth = 10$)
    if $|P| \leq 4$ or $depth > maxDepth$ **then**
        return BruteForceHamiltonianPath(P)
    end if
    $(a, b, c) \leftarrow$ FindMidpointLine(P)
    $(P_L, P_R) \leftarrow$ PartitionPoints(P, a, b, c)
    $\pi_L \leftarrow$ DivideAndConquerWiener(P_L, $depth + 1$)
    $\pi_R \leftarrow$ DivideAndConquerWiener(P_R, $depth + 1$)
    return OptimalConnectPaths(π_L, π_R)
end procedure

Divide and Conquer Algorithm for Approximating Hamiltonian Paths

Analyzing DivideAndConquerWiener proved to be very difficult. Initially, I thought it was a $\mathcal{O}(\log(n))$-Approximation but I could not show it.

I did additional empirical experiments with Python which you can find it here: github.com/Atishaya7777/spatially_embedded_graph

Hardness Proof

Euclidean Wiener Index Tree Problem (EWITP)

INSTANCE: A set P of n points in $\mathbb{R}^2$, a cost W, and a budget B.

QUESTION: Does there exist a spanning tree T of P such that $W(T) \leq T$ and $wt(T) \leq B$?

Here, $W(T)$ denotes the Wiener index of the tree T and $wt(T)$ denotes the total weight of the tree T.

Partition

INSTANCE: A set $X = \{x_1, x_2, \ldots, x_n\}$, where $x_i \in \mathbb{Z}$ such that $R = \sum_{i=1}^{n} x_i$.

QUESTION: Does there exists a subset $S \subseteq X$ such that $\sum_{x_i \in S} x_i = \frac{R}{2}$.

The following reduction is from [Abu-Affash et al., 2023].

Definition

Let P be a set of n points in $\mathbb{R}^2$ and let T be a spanning tree of P. For an edge $(p, q) \in T$, we define $N_T(p, q) = |\{x \in P : \delta_T(p, x) < \delta_T(q, x)\}|$ be the number of points in T that are closer to p than to q.

Hardness Proof

Given an instance $X = \{x_1, x_2, \ldots, x_n\}$ of the **PARTITION** instance where $x_i \in \mathbb{Z}$:

- Construct a set P of $m = n^3 + 3n + 1$ points.
  - A point s and n points $p_1, p_2, \ldots p_n$ located equally spaced on a circle of radius nR centered at s.
  - A cluster C of n^3 points located at the center of the circle of radius $\frac{1}{n^{10}R}$ centered at S.
  - For each $1 \leq i \leq n$, we have two points l_i and r_i both of distance x_i from p_i and the distance between them is $\frac{x_i}{2}$.
- Set

$$B = \left(n^2 + \frac{7}{4}\right)R + \frac{1}{n^7 R}$$

$$\begin{aligned} W &= 3n^2(m-3)R + \left(\frac{9}{4}m - \frac{13}{4}\right)R + \frac{1}{n^4 R} + \frac{3}{n^6 R} \\ &= 3n^5 R + \frac{45}{4}n^3 R - 6n^2 R + \frac{27}{4}nR - R + \frac{1}{n^4 R} + \frac{3}{n^6 R} \end{aligned}$$

Hardness Proof

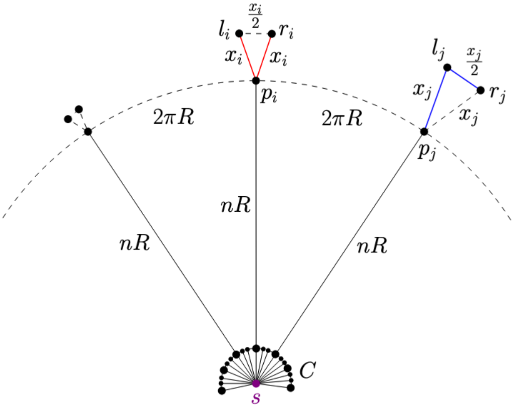

Figure 1: The set P produced by the reduction. Connecting the points l_j, r_j, and p_j for $x_j \in S$ (Blue) and connecting the points l_i, r_i, and p_i for $x_i \in X \setminus S$ (Red).

Hardness Proof

Assume there exists a set $S \subseteq X$ such that $\displaystyle\sum_{x_i \in S}^{x_i} = \frac{R}{2}$.

Construct a spanning tree T for the points of P as follows:

- $E_1 = \{(s, p_i) : 1 \leq i \leq n\}$
- $E_2 = \{(p_i, l_j), (p_i, r_i) : x_i \in S\}$
- $E_3 = \{(p_j, l_j), (l_j, r_j) : x_j \in X \setminus S\}$, and
- $E_4 = \{(s, p) : p \in C\}$

Weight of the Tree

$$\begin{aligned}
wt(T) &= wt(E_1) + wt(E_2) + wt(E_3) + wt(E_4) \\
&= n^2 R + R + \frac{3}{4}R + n^3 \cdot \frac{1}{n^{10}R} \\
&= \left(n^2 + \frac{7}{4}\right) R + \frac{1}{n^7 R} \\
&= B
\end{aligned}$$

Wiener Index of the Tree

$$
\begin{aligned}
W(T) &= \sum_{(p,q)\in T} N_T(p,q)\cdot N_T(q,p)\cdot |pq| \\
&= \sum_{(p,q)\in E_1} N_T(p,q)\cdot N_T(q,p)\cdot |pq| \\
&\quad + \sum_{(p,q)\in E_2} N_T(p,q)\cdot N_T(q,p)\cdot |pq| \\
&\quad + \sum_{(p,q)\in E_3} N_T(p,q)\cdot N_T(q,p)\cdot |pq| \\
&\quad + \sum_{(p,q)\in E_4} N_T(p,q)\cdot N_T(q,p)\cdot |pq| \\
&= W(\text{Chalk \& Talk})
\end{aligned}
$$

Converse direction

It is a matter of computation to show that:

Theorem (Claim 4)

There are exactly n edges in T between the points in $C \cup \{s\}$ and the others.

The paper shows it considering the cases of there are more than n edges and less than n edges and shows a contradiction.

Converse direction

Let P_i denote the triplet $\{p_i, l_i, r_i\}$ for $1 \leq i \leq n$.

- From Claim 4, it follows that for every $1 \leq i \leq n$, there is exactly one edge (p, q) in T for which $p \in C \cup \{s\}$ and $q \in P_i$. Note that $q = p_i$.
- Thus, in every P_i, we have $(p_i, l_i) \in T$ or $(p_i, r_i) \in T$.
- WLOG let $(p_i, l_i) \in T$:
  - Either $(p_i, r_i) \in T$ OR $(l_i, r_i) \in T$.
  - Let $S \subseteq X$ such that $x_i \in S \iff (p_i, r_i) \in T$.
  - Let $R' = \sum_{x_i \in S} x_i$.

They then argue that if $R' \neq \frac{R}{2}$, then either $wt(T) > B$ or $W(T) > W$.

This concludes the hardness result.

Why is the bound $wt(T) \leq B$ important?

The weight bound is what forces the gadgets to connect in a very specific way (open or closed), which in turn encodes the binary choices in **PARTITION**.

- n gadgest for each integer $x_i \in X$, giving two connection options:
  - **Open**: (p_i, l_i) and (p_i, r_i) contributing $2x_i$ weight.
  - **Closed**: (p_i, l_i) and (l_i, r_i) contributing $\frac{3x_i}{2}$ weight.

If we takeout the weight bound:

- We could just have everything open, which might reduce the Wiener index (Since open gadgets keep gadget paths short), but increase the weight
- In the unbounded case, we would have no reason to avoid such trees.

Next Steps and Open Problems?

- Preliminarily, I think the modification to the reduction needs to happen such that the penalty is no longer on the weight but solely on the Wiener Index. Of course, this is harder to prove.
- Primiarly, construct gadgets so that the only correct combination of open/closed configurations minimize the Wiener Index.

References I

Abu-Affash, A. K., Carmi, P., Luwisch, O., and Mitchell, J. S. B. (2023).
Geometric spanning trees minimizing the wiener index.

The end.

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